

Harmonised Survey Data: Technical Documentation

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Contents

1	Introduction	3
2	Data Collection	4
2.1	Sign Up SERL Observatory Survey – (2019–2021)	4
2.2	2023 SERL Observatory Survey – (2023)	4
2.3	2025 SERL Observatory Survey – (2025–2026)	4
3	Harmonised SERL Observatory Survey	6
3.1	Overview	6
3.2	Domain Structure	6
3.3	Harmonisation Methodology	6
3.3.1	Overview	6
3.3.2	Step 1 – Variable Identification and Mapping	7
3.3.3	Step 2 – Numeric Coding of Text Responses (2023 Survey)	8
3.3.4	Step 3 – Numeric Code Harmonisation	8
3.3.5	Step 4 – Construction of the MasterSERL Dataframe	8
3.3.6	Step 5 – Construction of the MasterSERL Dictionary	8
3.4	Assumptions & Simplifications	9
3.4.1	Thermostat data rules – Temperature Conversion	9
3.4.2	Questions with “added or replaced in last 12 months” options	9
3.4.3	Logic for blank answers for branching questions <code>stand_heater_own_</code> <code>power</code> and <code>stand_heater_use</code>	10
3.4.4	Logic for blank answers for branching questions <code>house_condition</code> and <code>mould_windows_bathroom</code> , <code>mould_walls</code> , and <code>mould_furnishing</code>	10

3.4.5	Logic for branching questions thermal_comfort / heating_difficulty_- home / heating_affordability / no_comfort_answer / none_above_comfort / other_reason_comfort	11
3.4.6	Question: no_occupants	11
3.4.7	Issues Relevant to Postal Survey Handling	11
3.4.8	Survey-Specific Questions	12
4	MasterSERL Dataset Dictionary	13

Introduction

This document describes the harmonised survey responses data table. This table combines variables from the Sign Up SERL Observatory survey, the 2023 SERL Observatory survey and the 2025 SERL Observatory survey, harmonised into a single analytical dataframe. The purpose of this dataset is to facilitate comparability across surveys and to support longitudinal analysis over time.

Variables are grouped by domain: Demographics, Energy, Appliances, Behaviour, Water, Housing, Environment, Finance, Comfort, and Health. Variables are drawn from original survey questions and have been harmonised to allow comparability across the three surveys.

Files currently available relating to the harmonised survey data are:

- `masterserl_surveys_edition{number}.csv`: the harmonised dataframe described in this document. ({number} is the number of the edition, e.g. "01")
- `master_serl_dictionary_edition{number}.xlsx`: a table matching the harmonised variables and response codes to the original survey questions and responses in .xlsx format.
- `master_serl_dictionary_edition{number}.csv`: a table matching the harmonised variables and response codes to the original survey questions and responses in .csv format.
- `serl_mastermapping_methodology_edition{number}.xlsx`: a spreadsheet with the methodology implemented for the harmonisation of the surveys.
- `serl_pilot_recruitment_survey_copy.pdf`: the postal survey sent to households in the first (pilot) recruitment wave.
- `serl_main_recruitment_survey_copy.pdf`: the postal survey sent to households in the main recruitment waves (very similar to the survey used in the pilot study).
- `serl_follow_up_survey_2023_copy.pdf`: the 2023 survey sent to households in the recruitment waves.
- `2025SERLSurveyFinalCopy.pdf`: the 2025 survey sent to households in the recruitment waves.

Data Collection

Sign Up SERL Observatory Survey – (2019–2021)

The Sign Up survey recruited participants in three waves, from 2019 until 2021.

The first wave was conducted during September and October 2019, and that was the SERL pilot study. Around 1,700 households signed up in wave 1 from 18,000 households that were invited to participate. The completion of the survey was not required for the households that consented, as opposed to the access to their smart meter data. Up to 4 letters were sent to these households, in central and southern England and in Wales, inviting them to participate. The participants had the option to sign-up online or complete the consent form and survey by post.

The second wave took place in September 2020, adding 3,200 participants, with same consent form to be completed. This time the households received up to 3 mailings inviting them to participate, with the first two offering only online sign-up, while the third offering both online and postal sign-up.

Wave 3 conducted in January 2021, recruiting approximately 8,000 additional participants, with participants encouraged to complete the survey, as in the previous waves. As for wave 2, households were sent up to 3 mailings, the first two offering online sign-up only and the third offering postal and online sign-up.

2023 SERL Observatory Survey – (2023)

The 2023 survey was sent SERL Observatory participants on 2 February 2023. The main purpose was to understand the way that the energy price rise in the UK has affected GB households in the winter of 2022/23. Thus, the SERL team developed a survey to assess energy use practices during this winter. In that survey 12,001 households were invited to participated by post, including a weblink for online survey completion, and from those 5,400 consented.

An overview of the findings from the survey can be found in [Self-reported energy use behaviour changed significantly during the cost-of-living crisis in winter 2022/23: insights from cross-sectional and longitudinal surveys in Great Britain](#).

2025 SERL Observatory Survey – (2025–2026)

The 2025 survey was conducted during March 2025 and April 2026. In particular, the 2025 survey was sent in three waves. Wave 1 was implemented from March 2025 to June 2025, and waves 2 and 3 were implemented from November 2025 to April 2026.

On 14 March 2025 Accent Market Research posted 11,586 annual reminder letters with a QR code linking to the online survey. From those, 1,944 households completed the survey, including 58 paper responses that were provided to the participants upon request. After that, reminders with printed copies of the survey were posted over the weekend of 25 April 2025 to 5,400 non-responders, selected if they responded to the 2023 Cost of Living survey but hadn't yet responded to the 2025 survey, and then based on the completeness of their electricity data to fill the budget available for a postal approach. In total, 3,806 participants responded during wave 1.

A further 470 participants responded to waves 2 and 3 of the survey, including (in error) 35 individuals who had previously participated in Wave 1. This time the survey was sent out via email link to any participants that consented to join the EDOL sample and have mini-meters in their homes. This was to boost numbers among those with sufficient smart meter data quality and to maintain representative-ness according to region and IMD strata.

Hence, 2025 survey recruited in total 4,241 participants.

Harmonised SERL Observatory Survey

Overview

The purpose of the SERL Observatory survey harmonisation is to allow users to conduct longitudinal analysis over time, in various research questions, aligning with the domains described in Domain Structure. It should be noted that there may be differences in numbers of participants in different editions of the SERL Observatory due to participant withdrawal with full erasure of data, which means that the number of rows of data of the current edition is likely to change in the future editions. The data has been processed to allow for as much comparability as possible between the different surveys.

Domain Structure

Variables are grouped into the following domains:

- **Demographics** – household composition, age, gender, employment, ethnicity, pets, occupancy
- **Energy** – heating systems, controls, upgrades, sustainable technologies, EV infrastructure
- **Appliances** – kitchen, laundry, cooling, transport, and smart home appliances
- **Behaviour** – energy-saving actions, heating habits, appliance usage patterns
- **Water** – methods of heating tap and shower water
- **Housing** – accommodation type, insulation, glazing, building age, mould
- **Environment** – thermal comfort, damp, outdoor space access, parking
- **Finance** – income, payment methods, fuel affordability
- **Comfort** – thermal comfort, reasons for not being warm
- **Health** – general health, long-term conditions, wellbeing scores
- **Other** – smart meter use, cars/vans, survey metadata

Harmonisation Methodology

Overview

The harmonisation of the three surveys, namely the Sign Up Survey (2019–2021), the 2023 Survey (2023), and the 2025 Survey (2025–2026) was conducted by manual Excel-based mapping and Python scripting. The purpose of the harmonisation was to produce a single dataframe (`MasterSERL_Surveys.csv`), called MasterSERL Observatory Survey data, that will allow for longitudinal and comparative analysis across all three surveys, while keeping the original data.

The final harmonised dataset contains **389 variables** divided into 11 domains based on their topic, drawn from the three SERL Observatory survey dataframes, which contained different variable names, response scales, and coding conventions. The domains and their variable counts are shown in Table 1. Variables not asked in a particular survey are coded as 999999, indicating that the question did not exist in that survey.

Table 1: Number of harmonised variables per domain.

Domain	Variables	Description
Behaviour	95	Occupant behaviour
Energy	88	Heating systems and energy infrastructure
Demographics	58	Household composition and participant characteristics
Appliances	37	Household electrical equipment
Housing	34	Building characteristics
Health	25	Physical and mental health
Environment	14	Building outdoor facilities
Water	12	Water heating and usage
Other	10	Miscellaneous variables
Finance	7	Financial questions
Comfort	5	Human thermal comfort
Total	385	

Step 1 – Variable Identification and Mapping

The first step involved a systematic identification of all variables across the three surveys and the construction of a Master Mapping table manually. For each original variable across the three surveys, the corresponding variable in the harmonised dataset was recorded. For instance, the answers in the variable `heating_gas` in the harmonised dataset, contained under different variable names for the three surveys, namely `A302` in the Sign Up Survey, `B2_2_yes` in the 2023 Survey, and `Q15_2` in the 2025 Survey. Also, where a variable did not exist in a given survey, the mapping was left blank.

Since the three SERL Observatory surveys were shaped by the socio-economic context at the time that each of them was sent to the participants, they were designed to address different research questions and interests. Therefore, there is a high number of variables that are unique to each survey. As shown in Table 2, the majority of variables (147 questions) appear only in the 2025 Survey, mirroring its wider scope, while only 27 variables are common to all three surveys.

Table 2: Distribution of harmonised variables by survey coverage.

Survey coverage	Number of variables
All three surveys (Sign Up, 2023, 2025)	27
Sign Up Survey only	54
2023 Survey only	40
2025 Survey only	147
Sign Up and 2023 Surveys	3
Sign Up and 2025 Surveys	32
2023 and 2025 Surveys	76
Total	379

Step 2 – Numeric Coding of Text Responses (2023 Survey)

In contrast to the Sign Up survey and the 2025 survey that stored the participant responses as numeric codes, the 2023 survey stored them as text strings (e.g. "Always work from home" rather than 1). The translation was carried out automatically, guided by a text-to-number mapping table (See AnswerMapping_Text tab in the `serl_mastermapping_methodology_edition01.xlsx`.)

Step 3 – Numeric Code Harmonisation

Each of the three surveys used different numeric coding schemes for equivalent response options. For example, a Yes/No question might use codes 1/0 in one survey and 1/2 in another. Thus, the following step was to standardise all numeric codes across surveys to a common harmonised coding scheme.

The harmonised coding types were defined in the AnswerMapping_Harmonisation mapping table in the `serl_mastermapping_methodology_edition01.xlsx`, which specified the target value and meaning for each response option. A Python script was then applied to recode all affected columns in each survey dataframe to the harmonised scheme prior to merging.

Step 4 – Construction of the MasterSERL Dataframe

The three survey dataframes, after their harmonisation process with the harmonised variable names and numeric codes, were merged into a single master dataframe (MasterSERL Dataframe) using `puprn` as the unique participant identifier. The merge was structured as a full outer join, preserving all participants from all three surveys. Variables not present in a given survey were filled with the value 999999, indicating that the question was not asked in that survey.

The resulting dataframe contains one row per participant per survey response, with all 389 harmonised variables as columns. The `survey` column identifies which survey each row originates from (Sign Up, 2023, 2025w_1, or 2025w_2_3).

Step 5 – Construction of the MasterSERL Dictionary

The final step involved creating the data dictionary (`MasterSERL_Dictionary.xlsx` and `MasterSERL_Dictionary.csv`) described in this document. For each of the 389 variables in the harmonised dataset, the dictionary records the domain, variable name, concept, all possible values, and

their meanings. Variables with multiple possible values are documented with one row per value. This allows the dictionary to serve as a complete reference for the harmonised MasterSERL dataset.

Assumptions & Simplifications

Thermostat data rules – Temperature Conversion

Any decimals in postal surveys were rounded to nearest whole number (e.g. 19.5 becomes 20, 19.1 becomes 19), only whole numbers could be entered online.

Where a range is apparently entered (e.g. thermostat_C = 2022) a midpoint is chosen (in this hypothetical case thermostat_C = 21 °C).

Where a decimal point seems to be missing this is corrected (e.g. thermostat_C = 205: thermostat_C = 20.5).

Rules that have been applied for the conversion of the degrees Fahrenheit (°F) to degrees Celsius (°C) are presented in Table 3.

Table 3: Temperature conversion rules applied to thermostat data.

Case	Action
thermostat_C in expected Celsius limits	No corrections made.
thermostat_C below lower Celsius limit	No corrections made.
thermostat_C above upper Celsius limit and below upper Fahrenheit limit: assumed to be °F	thermostat_C is value converted from F.
thermostat_C above upper Celsius limit but below lower Fahrenheit limit (i.e. between 35 and 50)	No corrections made.
thermostat_C above upper Fahrenheit limit	No corrections made.

Note: Celsius limits: 10 °C – 35 °C. Fahrenheit limits: 50 °F – 95 °F.

Questions with “added or replaced in last 12 months” options

Table 4 shows the logic for all questions where there is an added/replaced in last 12 months option, heating_gas and gas_add_rep_12m quoted as an example.

Table 4: Logic applied to “present” and “added/replaced in last 12 months” questions.

“present” column (e.g. heating_gas)	“added/replaced” column (e.g. heating_gas)	Action
“Yes”	“Yes”	No change.
“Yes”	“ ”	“added/replaced” column set to NA.
“ ”	“Yes”	“present” column set to “No response”.
“ ”	“ ”	“present” column set to “No”. “added/replaced” column set to “No”.

Logic for blank answers for branching questions *stand_heater_own_power* **and** *stand_heater_use*Table 5: Logic applied to blank answers for branching questions *stand_heater_own_power* and *stand_heater_use*.

<i>standalone_heater</i>	<i>stand_heater_own_power</i>	<i>stand_heater_use</i>	Action
" "	" "	" "	<i>stand_heater_own_power</i> coded "No response". <i>stand_heater_use</i> coded "No response".
No	" "	" "	<i>stand_heater_own_power</i> and <i>stand_heater_use</i> coded NA.
Yes	" "	" "	<i>stand_heater_own_power</i> coded "No response". <i>stand_heater_use</i> coded "No response".
Yes	"Powered by mains gas or electricity"	" "	<i>stand_heater_use</i> coded NA.
Yes	"Some or all have own source of fuel"	" "	<i>stand_heater_use</i> coded "No response".
Yes	"Some or all have own source of fuel"	One of 6 answers about frequency of use	No action required.

Note: standalone_heater: Do you use standalone heaters? | stand_heater_own_power: Any of your standalone heaters have their own source of power? | stand_heater_use: In very cold winter weather, how often are these standalone heaters typically used?

Logic for blank answers for branching questions *house_condition* **and** *mould_windows_bathroom, mould_walls, and mould_furnishing*

The question that has been asked for the *house_condition* variable is if the participants have problems with condensation, damp or mould in their house. Hence, the relation with the variables *mould_windows_bathroom, mould_walls, and mould_furnishing*.

Table 6: Logic applied to branching questions house_condition and mould variables.

house_condition	mould_windows_bathroom / mould_walls / mould_furnishing	Action
" "	" "	No action.
No	" "	mould_windows_bathroom etc coded NA.
Don't know	" "	mould_windows_bathroom etc coded NA.
Yes	All " "	mould_windows_bathroom etc coded "No response".
Yes	Any of possible answers marked to mould_windows_bathroom etc marked	All other answers to mould_windows_bathroom etc coded NA / "No".

Note: Some respondents ticked Don't know for house_condition and then selected one or more Don't know options for the mould variables – this means that the number of NA answers varies across mould_walls, mould_windows_bathroom, and mould_furnishing.

Logic for branching questions thermal_comfort / heating_difficulty_home / heating_affordability / no_comfort_answer / none_above_comfort / other_reason_comfort

Table 7: Logic applied to branching questions thermal_comfort and comfort variables.

thermal_comfort	Comfort Variables	Action
" "	" "	All comfort variables coded NA.
Yes	" "	All comfort variables coded NA.
Don't know	" "	All comfort variables coded NA.
No	" "	All comfort variables coded "No response".
No	One of 5 answers	All other comfort variables coded "No".

Note: Comfort variables: heating_difficulty_home / heating_affordability / no_comfort_answer / none_above_comfort / other_reason_comfort

Question: no_occupants

The answer to no_occupants is numeric, so no response is left as NA rather than set to "No response" which would change the data type of this column.

Issues Relevant to Postal Survey Handling

Several data quality issues specific to the postal survey format were identified and handled as follows:

- **Missing pages:** Where surveys were returned with missing pages, the relevant questions were left blank.

- **Multiple responses to single-answer questions:** Where a respondent marked more than one response to a question that required a single answer, no response was recorded.
- **Blank answers:** Where no box was ticked for a question in which an answer was expected, the blank response was recoded from "" to "No response".

Survey-Specific Questions

Questions that were not common across surveys have been retained in the dataset. Where a question was not included in a survey, the value 999999 was used to indicate that the question was not asked.

MasterSERL Dataset Dictionary

Table 8: Guide to the harmonised survey data: variables and concepts.

Variable name	Concept	Possible values		Value meanings	Domain
Other					
PUPRN	Unique respondent identifier	—	—		Other
survey	Survey information	Sign Up	—		Other
		2023	—		
		2025w_1	—		
		2025w_2_3	—		
		Most Recent Survey	Most recent survey for each PUPRN		
		Most Recent Answers	Most recent answer for each variable and each PUPRN		
completed_survey	Households that completed the survey	True False	— —		Other
date_completed	Date that survey has been completed - datetime format	—	—		Other
date_completed_string	Date that survey has been - text format	—	—		Other

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
wave	Wave information	su_ - Wave1	Sign Up - Wave 1	Other
		su_ - Wave2	Sign Up - Wave 2	
		su_ - Wave3	Sign Up - Wave 3	
		23_ - Wave1	2023 Survey- Wave 1	
		23_ - Wave2	2023 Survey- Wave 2	
		23_ - Wave3	2023 Survey- Wave 3	
		25_ - Wave1	2025 Survey- Wave 1	
		25_ - Wave2	2025 Survey- Wave 2	
		25_ - Wave3	2025 Survey- Wave 3	
smart_meter_own	Knowledge of having smart meter	-2	No answer	Other
		1	Yes	
		2	No	
smart_meter_use	Compared to before this winter, are you using your smart meter In-Home Display/Smart Energy Display?	1	More often	Other
		2	Less often	
		3	About the same	
		4	I don't have this	
		5	It is not working	
		6	Don't know	
		7	No response	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
no_cars_vans	How many cars and vans does your household own or have available to use?	-9	Not applicable	Other
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0	
		1	1	
		2	2	
		3	3 or more	
		4	Don't know	
no_cars_vans_prefernotsay	How many cars and vans does your household own or have available to use? - Prefer not to say	-9	Not applicable	Other
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
Demographics				
no_occupants	How many people live in household?	number of people	number of people	Demographics
male_0_15	Males in household (0-15)	number of people	number of people	Demographics
male_0_15	Males in household (0-15)	-9	Not applicable	Demographics
male_0_15	Males in household (0-15)	-3	Prefer not to say	Demographics
male_0_15	Males in household (0-15)	-2	No answer	Demographics
male_0_15	Males in household (0-15)	-1	Don't know	Demographics
male_16_24	Males in household (16-24)	number of people	number of people	Demographics
male_16_24	Males in household (16-24)	-9	Not applicable	Demographics

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
male_16_24	Males in household (16-24)	-3	Prefer not to say	Demographics
male_16_24	Males in household (16-24)	-2	No answer	Demographics
male_16_24	Males in household (16-24)	-1	Don't know	Demographics
male_25_44	Males in household (25-44)	number of people	number of people	Demographics
male_25_44	Males in household (25-44)	-9	Not applicable	Demographics
male_25_44	Males in household (25-44)	-3	Prefer not to say	Demographics
male_25_44	Males in household (25-44)	-2	No answer	Demographics
male_25_44	Males in household (25-44)	-1	Don't know	Demographics
male_45_64	Males in household (45-64)	number of people	number of people	Demographics
male_45_64	Males in household (45-64)	-9	Not applicable	Demographics
male_45_64	Males in household (45-64)	-3	Prefer not to say	Demographics
male_45_64	Males in household (45-64)	-2	No answer	Demographics
male_45_64	Males in household (45-64)	-1	Don't know	Demographics
male_65_74	Males in household (65-74)	number of people	number of people	Demographics
male_65_74	Males in household (65-74)	-9	Not applicable	Demographics
male_65_74	Males in household (65-74)	-3	Prefer not to say	Demographics

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
male_65_74	Males in household (65-74)	-2	No answer	Demographics
male_65_74	Males in household (65-74)	-1	Don't know	Demographics
male_75_84	Males in household (75-84)	number of people	number of people	Demographics
male_75_84	Males in household (75-84)	-9	Not applicable	Demographics
male_75_84	Males in household (75-84)	-3	Prefer not to say	Demographics
male_75_84	Males in household (75-84)	-2	No answer	Demographics
male_75_84	Males in household (75-84)	-1	Don't know	Demographics
male_85plus	Males in household (85+)	number of people	number of people	Demographics
		-9	Not applicable	
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
female_0_15	Females in household (0-15)	number of people	number of people	Demographics
female_0_15	Females in household (0-15)	-9	Not applicable	Demographics
female_0_15	Females in household (0-15)	-3	Prefer not to say	Demographics
female_0_15	Females in household (0-15)	-2	No answer	Demographics
female_0_15	Females in household (0-15)	-1	Don't know	Demographics

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
female_16_24	Females in household (16-24)	number of people	number of people	Demographics
female_16_24	Females in household (16-24)	-9	Not applicable	Demographics
female_16_24	Females in household (16-24)	-3	Prefer not to say	Demographics
female_16_24	Females in household (16-24)	-2	No answer	Demographics
female_16_24	Females in household (16-24)	-1	Don't know	Demographics
female_25_44	Females in household (25-44)	number of people	number of people	Demographics
female_25_44	Females in household (25-44)	-9	Not applicable	Demographics
female_25_44	Females in household (25-44)	-3	Prefer not to say	Demographics
female_25_44	Females in household (25-44)	-2	No answer	Demographics
female_25_44	Females in household (25-44)	-1	Don't know	Demographics
female_45_64	Females in household (45-64)	number of people	number of people	Demographics
female_45_64	Females in household (45-64)	-9	Not applicable	Demographics
female_45_64	Females in household (45-64)	-3	Prefer not to say	Demographics
female_45_64	Females in household (45-64)	-2	No answer	Demographics
female_45_64	Females in household (45-64)	-1	Don't know	Demographics

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
female_65_74	Females in household (65-74)	number of people	number of people	Demographics
female_65_74	Females in household (65-74)	-9	Not applicable	Demographics
female_65_74	Females in household (65-74)	-3	Prefer not to say	Demographics
female_65_74	Females in household (65-74)	-2	No answer	Demographics
female_65_74	Females in household (65-74)	-1	Don't know	Demographics
female_75_84	Females in household (75-84)	number of people	number of people	Demographics
female_75_84	Females in household (75-84)	-9	Not applicable	Demographics
female_75_84	Females in household (75-84)	-3	Prefer not to say	Demographics
female_75_84	Females in household (75-84)	-2	No answer	Demographics
female_75_84	Females in household (75-84)	-1	Don't know	Demographics
female_85plus	Females in household (85+)	number of people	number of people	Demographics
		-9	Not applicable	
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
no_occupants_under_16	Number of occupants aged under 16	number of people	number of people	Demographics
no_occupants_over_16	Number of occupants aged 16+	number of people	number of people	Demographics
no_occupants_over_65	Number of occupants aged 65+	number of people	number of people	Demographics

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
no_occupants_over_65_all	All occupants aged 65+	-2	No answer	Demographics
work_hours_30plus	Working Status - Working 30 + hours per week	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0 people	
		1	1 person	
		2	2 people	
		3	3 people	
		4	4 or more people	
work_hours_30minus	Working Status - Working less than 30 hours per week	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0 people	
		1	1 person	
		2	2 people	
		3	3 people	
		4	4 or more people	
no_work_disable	Working Status - Not Working - long term sickness or disability	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0 people	
		1	1 person	
		2	2 people	
		3	3 people	
		4	4 or more people	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
unemployed	Working Status - Unemployed but seeking work	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0 people	
		1	1 person	
		2	2 people	
		3	3 people	
		4	4 or more people	
student	Working Status - Student	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0 people	
		1	1 person	
		2	2 people	
		3	3 people	
		4	4 or more people	
retired_WFH	Working Status - Retired / at home / not seeking work	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0 people	
		1	1 person	
		2	2 people	
		3	3 people	
		4	4 or more people	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
work_status_ other	Working Status - Other	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0 people	
		1	1 person	
		2	2 people	
		3	3 people	
		4	4 or more people	
none_working	No adults work	0	No	Demographics
		1	Yes	
degree_holders	How many people in household hold a degree or higher qualification?	-9	Not applicable	Demographics
		-3	Don't know/ Prefer not to say	
		-2	No answer	
degree_ prefernosay	How many people in household hold a degree or higher qualification? - Prefer not to say	number of people	number of people	Demographics
age_group	Age Group	-2	No answer	Demographics
		1	Under 18	
		2	18-24	
		3	25-44	
		4	45-64	
		5	65-74	
		6	75-84	
		7	85+	
gender	Gender	-3	Prefer not to say	Demographics
		-2	No answer	
		1	Male	
		2	Female	
		3	In some other way	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
current_employment_status	Current Employment Status	-3	Prefer not to say	Demographics
		-2	No answer	
		1	Working (paid or unpaid): 30 hours a week or more	
		2	Working (paid or unpaid): less than 30 hours a week	
		3	Not working because of long-term sickness or disability	
		4	Unemployed but seeking work	
		5	A student	
age_group_0_5	How many people are there in 0-5 age group in your household?	6	Retired/at home/not seeking work (including looking after the home or family)	Demographics
		7	Other	
		0	0 people	
		1	1 person	
		2	2 people	
age_group_0_6	How many people are there in 0-5 age group in your household?	3	3 people	Demographics
age_group_0_7	How many people are there in 0-5 age group in your household?	4	4 or more people	Demographics

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
age_group_0_10	How many people are there in 0-5 age group in your household?	5	No response	Demographics
age_group_6_15	How many people are there in 6-15 age group in your household?	0	0 people	Demographics
age_group_6_16	How many people are there in 6-15 age group in your household?	1	1 person	Demographics
age_group_6_17	How many people are there in 6-15 age group in your household?	2	2 people	Demographics
age_group_6_18	How many people are there in 6-15 age group in your household?	3	3 people	Demographics
age_group_6_19	How many people are there in 6-15 age group in your household?	4	4 or more people	Demographics
age_group_6_20	How many people are there in 6-15 age group in your household?	5	No response	Demographics
age_group_0_15	How many people are there in 0-15 age group in your household?	number of people	number of people	Demographics
age_group_16_24	How many people are there in 16-24 age group in your household?	number of people	number of people	Demographics
age_group_25_44	How many people are there in 24-44 age group in your household?	number of people	number of people	Demographics

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
age_group_45_64	How many people are there in 45-64 age group in your household?	number of people	number of people	Demographics
age_group_65_74	How many people are there in 65-74 age group in your household?	number of people	number of people	Demographics
age_group_75_84	How many people are there in 75-85 age group in your household?	number of people	number of people	Demographics
age_group_85plus	How many people are there over 85 years old in your household?	number of people	number of people	Demographics
WFH_situation	Does anyone in your household work in the following ways? Always work from home; Sometimes work from home; Never work from home; Not applicable /prefer not to say	1 2 3 4 5	Always work from home Sometimes work from home Never work from home Not applicable /prefer not to say No response	Demographics
weekday_occupancy	Is there usually at least one person in the home on a weekday?	-9 -3 -2 -1 1 2 3	Not applicable Prefer not to say No answer Don't know Yes No It varies a lot	Demographics

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
weekday_occ_frequency	How often is at least one adult in your household working or engaging in other activities (e.g. childcare) from home on a weekday?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	3 or more times a week	
		2	Once or twice a week	
		3	Less than that but more than twice a month	
		4	Once or twice a month	
		5	Less than that but more than twice a year	
no_pets	Do you have any pets that live with your household? No	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
dog	Do you have dog?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
cat	Do you have cat?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
rabbit	Do you have rabbit?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
indoor_bird	Do you have indoor bird?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
guinea_pigs	Do you have guinea pigs?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
hamsters	Do you have hamsters?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
tortoises_turtles	Do you have tortoises and turtles?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
lizard	Do you have lizard?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
snake	Do you have snake?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
domestic_fowl	Do you have domestic fowl?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
indoor_fish	Do you have indoor fish?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
other_pets	Do you have other pets?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
ethnic_group	To which ethnic group do you belong?	-9	Not applicable	Demographics
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Indian	
		2	Pakistani	
		3	Bangladeshi	
		4	Chinese	
		5	Any other Asian back-ground	
		6	Caribbean	
		7	African	
		8	Any other Black, Black British, or Caribbean background	
		9	White and Black Caribbean	
		10	White and Black African	
		11	White and Asian	
		12	Any other Mixed or multiple ethnic back-ground	
		13	English, Welsh, Scottish, Northern Irish or British	
		14	Irish	
		15	Gypsy or Irish Traveller	
		16	Roma	
		17	Any other White back-ground	
		18	Arab	
		19	Any other ethnic group	
		20	Prefer not to say	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
heating_type_ none	No central heating	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
heating_gas	Gas heating present	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
heating_el_ storage	Electric storage heaters present	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
heating_el_ radiators	Electric radiators present	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
heating_other_ elec	Other electric heating present	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
heating_oil	Oil heating present	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
heating_soild_fuel	Solid fuel present	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
heating_biomass	Biomass present	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
heating_district	District or community present	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
heating_other	Other heating present	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
heating_timer	Central heating controls-timer	-9	Not applicable	Energy
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
heating_temp_set	Central heating controls-temperature setting	-9 0 1	Not applicable No Yes	Energy
heating_smart_dev	Central heating controls-smart device	-9 0 1	Not applicable No Yes	Energy
heating_manually	Central heating controls-manually	-9 0 1	Not applicable No Yes	Energy
heating_no_controls	Central heating controls-none of the above	-9 0 1	Not applicable No Yes	Energy
dk_heating_control	Central heating controls-don't know	-9 0 1	Not applicable No Yes	Energy
thermostat_C	Temperature controller set at for late afternoons and evenings_Degrees C	real temperature/number	real temperature/number	Energy
heating_upgrade_no	Considering replacing or adding heating or energy-Not considering	0 1	No Yes	Energy
heating_upgrade_na	Considering replacing or adding heating or energy-Not applicable	0 1	No Yes	Energy
heating_upgrade_GB	Considering replacing or adding heating or energy-Gas Boiler	0 1	No Yes	Energy
heating_upgrade_EL	Considering replacing or adding heating or energy-Solar panels for electricity	0 1	No Yes	Energy

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
heating_upgrade_SW	Considering replacing or adding heating or energy-Solar water heating	0	No	Energy
		1	Yes	
heating_upgrade_FB	Considering replacing or adding heating or energy-Wood or fuel burner	0	No	Energy
		1	Yes	
heating_upgrade_HP	Considering replacing or adding heating or energy-Heat pump	0	No	Energy
		1	Yes	
heating_upgrade_HPU	Considering replacing or adding heating or energy-Micro combined HPU	0	No	Energy
		1	Yes	
heating_upgrade_WT	Considering replacing or adding heating or energy-Wind turbine	0	No	Energy
		1	Yes	
heating_upgrade_other	Considering replacing or adding heating or energy-other specify	0	No	Energy
		1	Yes	
heating_heatpump	heat pump heating present	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
heating_add_rep_12m	Has a heating system been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
gas_add_rep_12m	Has a gas boiler been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
el_storage_add_rep_12m	Have electric storage heaters been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
el_radiators_add_rep_12m	Have electric radiators been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
heatpump_add_rep_12m	Has a heat pump been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
other_el_heat_add_rep_12m	Has another electric heating system been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
oil_add_rep_12m	Has an oil heating system been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
solidfuel_add_rep_12m	Has an solid fuel heating system been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
biomass_add_rep_12m	Has a biomass boiler been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
district_add_rep_12m	Has district or community heating system been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
other_add_rep_12m	Has other heating system been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
solar_panels_electricity	Does your household have solar panels for electricity?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
solar_water_heating	Does your household have solar water heating?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
battery_storage	Does your household have battery storage that is wired into your mains electricity?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
EV_point	Does your household have electric vehicle (EV) charging point?	1 0 -1 -2 -3	Yes No Not applicable No response Don't know	Energy
LED_lights	Does your household have LED light bulbs in most of your lights?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Energy
radiators_main_living	Does your household have thermostatic radiator valves on at least half of your radiators?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Energy
TRVs_present	Does your household have radiators in main living areas?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Energy
UFH_main_living	Does your household have underfloor heating in main living areas?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Energy

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
air_heating_main_living	Does your household have warm air heating units in main living areas?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
thermostat_present	Does your household have central heating thermostat?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
time_clock_present	Does your household have central heating time clock / programmer?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
smart_control_present	Does your household have smart heating control (e.g., Nest, Hive, Tado)?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
HW_cylinder_present	Does your household have hot water cylinder?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
HWC_insulation_present	Does your household have insulating jacket on hot water cylinder?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
HWC_thermostat_present	Does your household have thermostat on hot water cylinder?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
solar_panels_add_rep_12m	Have solar panels for electricity been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
solarwater_add_rep_12m	Has solar water heating been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
bat_storage_add_rep_12m	Has battery storage that is wired into your mains electricity been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
EV_point_add_rep_12m	Has electric vehicle (EV) charging point been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Energy
		2	No	
LED_lights_add_rep_12m	Have LED light bulbs in most of your lights been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
radiators_add_rep_12m	Have radiators in main living areas been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
TRVs_add_rep_12m	Have thermostatic radiator valves on at least half of your radiators been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Energy
		2	No	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
UFH_add_rep_12m	Has underfloor heating in main living areas been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
air_heaters_add_rep_12m	Have warm air heating units in main living areas been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
thermostat_add_rep_12m	Has central heating thermostat been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Energy
		2	No	
time_clock_add_rep_12m	Has central heating time clock / programmer been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Energy
		2	No	
smart_control_add_rep_12m	Has smart heating control (e.g., Nest, Hive, Tado) been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Energy
		2	No	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
HWC_add_rep_12m	Has hot water cylinder been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
HWC_ins_add_rep_12m	Have insulating jacket on hot water cylinder been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
HWC_thermostat_add_rep_12m	Has thermostat on hot water cylinder been added or replaced in your home in the last 12 months?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes, Has been added or replaced in the last 12 months	
none_sustain_add_rep_12m	Has another sustainable metre been added or replaced in your home in the last 12 months? - None of the above	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
dk_sustain_add_rep_12m	Has another sustainable metre been added or replaced in your home in the last 12 months? - Don't know	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
no_EVs	How many plug-in electric vehicles does your household have?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0	
		1	1	
		2	2	
no_EVs_12m	How many of the EVs has your household acquired in the last 12 months?	3	3 or more	Energy
		0	0	
		1	1	
		2	2	
		3	3 or more	
		4	Don't know	
		5	No response	
no_sustainable_present	Does your household have other sustainable solutions? No sustainable solution	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
dk_sustainable_present	Does your household have other sustainable solutions? Don't know	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
house_heat_zones	Can different zones of the house be independently heated to different temperatures at different times?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
boiler_flow_temp_adjust	Can the boiler flow temperature be adjusted?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
no_heat_controls	Does your household have heating controls? No	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
dk_heat_controls	Does your household have heating controls? Don't know	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
EV_charge_location	Where does your household mostly charge your electric vehicle(s)?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	At home, using a three-pin plug (trickle charge)	
		2	At home, using a dedicated EV charger	
		3	Near home, using an EV charger provided by third party for residents	
		4	At work, at a charging station provided by my workplace	
		5	At public charging stations (e.g. at supermarkets, service stations, public car parks)	
thermostat_use	Do you use the central heating thermostat?	6	Somewhere else	Energy
		7	Don't know	
		-9	Not applicable	
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
time_clock_use	Do you use the central heating time clock/ programmer?	0	No	Energy
		1	Yes	
		-9	Not applicable	
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
smart_control_use	Do you use the smart heating control (e.g., Nest, Hive, Tado)	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
TRVs_use	Do you use the thermostatic radiator valves (TRVs) on more than half of radiators?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
house_heat_zones_use	Do you use different zones of the house that can independently heated to different temperatures at different time?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
boiler_flow_temp_use	Do you use the boiler flow temperature that can be adjusted?	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
no_heat_controls_use	You don't use none of the above.	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
dk_use_of_heat_controls	You don't know.	-9	Not applicable	Energy
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
Appliances				
standalone_heater	Use of standalone heaters	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
stand_heater_own_power	Any of your standalone heaters have their own source of power	-9	Not applicable	Appliances
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	
stand_heater_use	In very cold winter weather, how often are these standalone heaters typically used?	-9	Not applicable	Appliances
		-2	No answer	
		-1	Don't know	
		1	Daily	
		2	Most days	
		3	Rarely - only if I really have to	
		4	Never	
		5	Varies, depends on temp or other reasons	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
electric_oven	Appliances in accommodation - Kitchen - Electric Oven	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
gas_oven	Appliances in accommodation - Kitchen - Gas Oven	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
electric_hob	Appliances in accommodation - Kitchen - Electric Hob	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
gas_hob	Appliances in accommodation - Kitchen - Gas Hob	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
dishwasher	Appliances in accommodation - Kitchen - Dishwasher	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
fridge_freezer	Appliances in accommodation - Kitchen - Fridge or fridge-freezer	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
freezer	Appliances in accommodation - Kitchen - Separate stand alone freezer	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
washer_dryer	Appliances in accommodation - Laundry - Combined clothes washer dryer	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
washer	Appliances in accommodation - Laundry - Washing machine	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
dryer	Appliances in accommodation - Laundry - Tumble dryer	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
laptop_PC	Appliances in accommodation - Consumer electronics - Laptop, computer	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
TV	Appliances in accommodation - Consumer electronics - TV	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
AC	Appliances in accommodation - Cooling - air conditioning unit	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
cooling_fan	Appliances in accommodation - Cooling - cooling fan	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
EV_present	Does household have plug in electric vehicle?	-2	No answer	Appliances
		-1	Don't know	
		1	Yes	
		2	No	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
el_induction_hob	Appliances in accommodation - Kitchen - Electric Induction Hob	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
other_kitchen_appliance	Appliances in accommodation - Kitchen - Other	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
microwave	Appliances in accommodation - Kitchen - Microwave	0	No	Appliances
		1	Yes	
air_fryer	Appliances in accommodation - Kitchen - Air fryer	0	No	Appliances
		1	Yes	
slow_cooker	Appliances in accommodation - Kitchen - Slow cooker	0	No	Appliances
		1	Yes	
smart_plug	Appliances in accommodation - Kitchen - Smart plug	0	No	Appliances
		1	Yes	
smart_thermostat	Appliances in accommodation - Kitchen - Smart thermostat	0	No	Appliances
		1	Yes	
el_drying_rack	Appliances in accommodation - Laundry - Electric Drying Rack (heated airer)	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
iron	Appliances in accommodation - Laundry - Electric Iron	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
vacuum_cleaner	Appliances in accommodation - Housework - Vacuum Cleaner	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
electric_shower	Appliances in accommodation - Electric Shower	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
smart_speakers	Appliances in accommodation - Smart home assistants / speakers (e.g. Alexa)	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
other_smart_eqp	Appliances in accommodation - Other smart home solutions, like smart plugs or app enabled appliances:	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
dehumidifier	Appliances in accommodation - Air quality - Dehumidifier	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
humidifier	Appliances in accommodation - Air quality - Humidifier	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
electric_heater	Appliances in accommodation - Heating - Electric Heater	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
e_scooter	Appliances in accommodation - Transport - E-scooter	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
e_bike	Appliances in accommodation - Transport - E-bike	-9	Not applicable	Appliances
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
e_cargo_bike	Appliances in accommodation - Transport - E-cargo bike	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Appliances
Behaviour				
el_gas_use_beh	Electricity or gas use behaviour	-9 -2 -1 1 2 3 4	Not applicable No answer Don't know A lot Some A little Not at all	Behaviour
heat_adjust_cold	Do you adjust your heating - When especially cold	0 1	No Yes	Behaviour
heat_adjust_baby	Do you adjust your heating - Because of children, infants, babies	0 1	No Yes	Behaviour
heat_adjust_visitors	Do you adjust your heating - Visitors	0 1	No Yes	Behaviour
heat_adjust_pets	Do you adjust your heating - Pets	0 1	No Yes	Behaviour
heat_adjust_stress	Do you adjust your heating - Stress	0 1	No Yes	Behaviour
heat_adjust_WFH	Do you adjust your heating - Working from home	0 1	No Yes	Behaviour
heat_adjust_none	Do you adjust your heating - none of these	0 1	No Yes	Behaviour

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
heat_adjust_unocc	how do you adjust heating controls when house is unoccupied	-2	No answer	Behaviour
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	
		6	Not applicable / Cannot do this	
unheated_spaces	During the winter, are there living spaces not normally heated?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
switch_off_lights_freq	How often do you .. Switch off lights	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	
		-9	Not applicable, cannot do this	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
clothes_freq	How often do you .. Put on more clothes when cold	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	
		-9	Not applicable, cannot do this	
energy_save_effort	How much effort do you say your household makes to limit energy usage	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	A great deal of effort	
		2	Some effort	
		3	A little effort	
		4	No effort at all	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
open_window_cold	How often will your household open the windows .. on a typical cold day	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Open them all the time/ Always	
		2	Open them a few times a day (e.g. an hour in the morning and an hour in the evening) / Very often	
		3	Open them once a day for a limited period (e.g. first thing in the morning) / Quite often	
		4	Open them a few times a week (e.g. if it gets hot /stuffy) / Not very often	
open_window_warm	How often will your household open the windows .. on a typical warm day	5	Never open them / Never	Behaviour
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
EV_charge_time	When does household charge electric vehicle at home?	-9	Not applicable, charging is not done at home (e.g. at work)	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Mostly overnight (8pm to 7am)	
		2	Mostly during the day (7am to 4pm)	
		3	Mostly in the evening (4pm to 8pm)	
		4	Varies - whenever it's needed	
boiler_flow_temp	During this winter, have you reduced the flow temperature of your boiler?	1	Yes	Behaviour
		0	No	
		-1	Not applicable	
		-2	No response	
		-3	Don't know	
heat_less_hours	During this winter, have you heated your house for fewer hours than in previous winters?	1	Yes	Behaviour
		0	No	
		-1	Not applicable	
		-2	No response	
		-3	Don't know	
vacant_heatoff	Turn down your thermostat or turn off your heating when you leave the house	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
standalone_ heatoff	Use a standalone heater rather than putting the heating on or turning it up	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
elect_blanket_ use	Use an electric blanket, hot water bottle or similar when feeling cold rather than putting the heating on or turning it up	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
vacant_room_ heatoff	Turn down radiators when rooms are not in use	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
occupied_room_ heatoff	Turn down radiators in rooms that are in use	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	
full_load_ washer	Run the washing machine with a full load rather than running it part full	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	
washer_30C	Run the washing machine at 30 degrees or lower	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
no_dryer_use	Dry clothes without using a tumble dryer	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	
standby_off	Turn appliances off standby when they are not in use	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	
curtains_blinds	Close curtains / blinds at night	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
shower_no_bath	Take a shower rather than having a bath	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
short_showers_no_long	Take short showers rather than longer showers	5	Never	Behaviour
		-9	Not applicable, cannot do this	
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
dishwasher_use_freq	How often do you use the dishwasher rather than washing up?	4	Not very often	Behaviour
		5	Never	
		6	Not applicable	
		7	cannot do this	
		8	Not applicable, cannot do this	
		9	No response	
		1	Always	
		2	Very often	
		3	Quite often	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
avoid_cooker_oven	Avoid using the cooker or oven when preparing a main meal	-9	Not applicable, cannot do this	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	
lights_off_compare	Switched off lights in rooms that aren't being used - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	
clothes_on_compare	Put more clothes on when feeling cold rather than putting the heating on or turning it up - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
heat_off_compare	Turned down your thermostat or turn off your heating when you leave the house - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	
standalone_compare	Used a standalone heater rather than putting the heating on or turning it up - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	
el_blanket_compare	Used an electric blanket, hot water bottle or similar when feeling cold rather than putting the heating on or turning it up - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
vacant_heatoff_compare	Turned down radiators when rooms are not in use - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	
occupied_heatoff_compare	Turned down radiators in rooms that are in use - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	
full_load_compare	Ran the washing machine with a full load rather than running it part full - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
compare_30C	Ran the washing machine at 30 degrees or lower - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	
no_dryer_compare	Dried clothes without using a tumble dryer - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	
standby_off_compare	Turned appliances off standby when they are not in use - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
curtains_ blinds_compare	Closed curtains / blinds at night - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	
shower_no_bath_ compare	Taken a shower rather than having a bath - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	
short_showers_ compare	Taken short showers rather than longer showers - Compare to last winter	1	A lot more	Behaviour
		2	A little more	
		3	About the same	
		4	A little less	
		5	A lot less	
		6	Not applicable	
		7	cannot do this	
		8	No response	
		9	Not applicable, cannot do this	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
dishwasher_use_compare	Used the dishwasher rather than washing up - Compare to last winter	1 2 3 4 5 6 7 8 9	A lot more A little more About the same A little less A lot less Not applicable cannot do this No response Not applicable, cannot do this	Behaviour
avoid_cooker_oven_compare	Avoided using the cooker or oven when preparing a main meal - Compare to last winter	1 2 3 4 5 6 7 8 9	A lot more A little more About the same A little less A lot less Not applicable cannot do this No response Not applicable, cannot do this	Behaviour
electric_oven_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the electric oven at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
gas_oven_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the gas oven at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
electric_hob_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the electric hob at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
el_induction_hob_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the electric induction hob at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
gas_hob_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the gas hob at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
other_kitchen_apps_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use other kitchen appliances (e.g. Microwave/ Air fryer/ Instant pot) at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
dishwasher_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the dishwasher at a different time of day compared to weekdays when you are away from home?	1	Always	Behaviour
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	
		6	Not applicable	
		7	cannot do this	
		8	Not applicable, cannot do this	
		9	No response	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
fridge_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the fridge or combined fridge freezer at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
freezer_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the separate stand-alone freezer at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
washer_dryer_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the combined clothes washer dryer at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
washer_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the washing machine at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
dryer_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the tumble dryer at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
heated_airer_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the electric drying rack (heated airer) at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
iron_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the iron at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
vacuum_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the vacuum cleaner at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
el_shower_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the electric shower at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
PC_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the laptop/computer(s) at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
TV_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the TV at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
smart_speakers_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the smart home assistants / speakers (e.g. Alexa) at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
other_device_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use other smart home solutions, like smart plugs or app enabled appliances, at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
aircon_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the air conditioning unit at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
cool_fan_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the cooling fan at a different time of day compared to weekdays when you are away from home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
dehumidifier_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the dehumidifier at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
humidifier_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the humidifier at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
el_heater_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the electric heater at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
e_scooter_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the electric heater at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
e_bike_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the e-bike at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
e_cargo_bike_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the e-cargo bike at a different time of day compared to weekdays when you are away from home?	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
no_diff_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use or charge any of these appliances at a different time of day compared to weekdays when you are away from home? No	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Behaviour
microwave_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the microwave at a different time of day compared to weekdays when you are away from home?	0 1	No Yes	Behaviour
air_fryer_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the air fryer at a different time of day compared to weekdays when you are away from home?	0 1	No Yes	Behaviour

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
slow_cooker_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the slow cooker at a different time of day compared to weekdays when you are away from home?	0 1	No Yes	Behaviour
smart_plug_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the smart plug at a different time of day compared to weekdays when you are away from home?	0 1	No Yes	Behaviour
smart_thermostat_use	When you are working or engaging in other activities (e.g. childcare) from home on weekdays, do you tend to use the smart thermostat at a different time of day compared to weekdays when you are away from home?	0 1	No Yes	Behaviour

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
smart_display_ use_freq	How often, if at all, do you or anyone in your household look at information about how much energy you are using (or your credit balance) through your smart meter In-Home Display/Smart Energy Display?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Most days	
		2	About once or twice per week	
		3	About once or twice per month	
		4	Less often than once per month	
		5	Never	
heating_hours	During the winter months, for how many hours a day do you typically heat your home?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
dk_heating_ hours	During the winter months, for how many hours a day do you typically heat your home? - Don't know	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
heat_off_ unoccupied	When your home is unoccupied for a day or more, turn the heating down or off?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Very often	
		3	Quite often	
		4	Not very often	
		5	Never	
		6	Not applicable, cannot do this	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
summer_windows	In the summer, do you open the windows to keep the home cool?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Not applicable, cannot do this	
summer_fans	In the summer, do you switch on fan(s) to keep the home cool?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Not applicable, cannot do this	
summer_shutters	In the summer, do you close shutters to keep the home cool?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Not applicable, cannot do this	
summer_curtains	In the summer, do you close curtains to keep the home cool?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Not applicable, cannot do this	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
summer_blinds	In the summer, do you close blinds to keep the home cool?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Not applicable, cannot do this	
summer_awning	In the summer, do you unroll an awning or canopy to keep the home cool?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Not applicable, cannot do this	
summer_aircon	In the summer, do you switch on air conditioner to keep the home cool?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Not applicable, cannot do this	
summer_MVHR	In the summer, do you use Mechanical Extract Ventilation or Mechanical Ventilation with Heat Recovery systems to keep the home cool?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Not applicable, cannot do this	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
summer_night_windows	In the summer, are you able to keep yourself cool at night by opening the windows?	-9	Not applicable	Behaviour
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Always	
		2	Often	
		3	Sometimes	
		4	Never	
		5	My home does not over-heat in summer / I can keep cool at night without opening windows	
Water				
tap_gas	Tap water heated (Gas Boiler)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	
tap_electric	Tap water heated (Electric Heater)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
tap_solar	Tap water heated (Solar Water Heater)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	
tap_other	Tap water heated (Other)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	
no_tap_heat	Tap water heated (Not Applicable)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	
dk_tap_heat	Tap water heated (Don't know)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
shower_gas	Shower water heated (Gas Boiler)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	
shower_electric	Shower water heated (Electric Heater)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	
shower_solar	Shower water heated (Solar Water Heater)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	
shower_other	Shower water heated (Other)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
no_shower_heat	Shower water heated (Not Applicable)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	
dk_shower_heat	Shower water heated (Don't know)	-9	Not applicable	Water
		-3	Prefer no say	
		-2	No answer	
		-1	Don't know	
		1	Yes, own source power	
		2	No, all are powered by mains gas and el.	
Housing				
accom_type	What type of accommodation do you live in	-2	No answer	Housing
		1	detached	
		2	semi detached	
		3	terraced (including end terrace)	
		4	in a purpose-built block of flats or tenement	
		5	part of a converted or shared house (including bedsits)	
		6	in a commercial building (e.g. in an office building, hotel or over a shop)	
self_contained	Is your accommodation self contained?	-2	No answer	Housing
		1	Yes, all the rooms are behind a door that only my household can use	
		2	No	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
no_house_share	How many other households do you share with?	-9	Not applicable	Housing
		-2	No answer	
own_or_rent	Do you own or rent this accommodation	-2	No answer	Housing
		1	Own it outright / buying it with a mortgage/loan	
		2	Part own and part rent (shared ownership)	
		3	Privately	
		4	From council (local authority) or housing association	
		5	Live here rent free	
no_rooms	How many rooms are available for use by this household?	-2	No answer	Housing
		-4	Unusable	
no_bedrooms	How many of these rooms count as bedrooms	-2	No answer	Housing
		-4	Unusable	
house_age	When was your accommodation built?	-2	No answer	Housing
		-1	Don't know	
		1	Before 1900	
		2	1900 to 1929	
		3	1930 to 1949	
		4	1950 to 1975	
		5	1976 to 1990	
		6	1991 to 2002	
		7	2003 onwards	
no_bathrooms	number of bathrooms	0	0	Housing
		1	1	
		2	2	
		3	3	
		4	4 or more	
		5	No response	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
loft_insulation	Does your household have loft insulation?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
cavity_wall_insulation	Does your household have cavity wall insulation?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
solid_wall_insulation	Does your household have solid wall insulation?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
underfloor_insulation	Does your household have underfloor insulation?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
double_glazing	Does your household have double or secondary glazed windows?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
draughtproofing	Does your household have draughtproofing on windows, doors and/or chimneys?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
other_insulation	Other type of insulation?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
dk_insulation	Don't know if house has any insulation.	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
no_insulation	None of the above types of insulation.	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
loft_ins_add_rep_12m	Has loft insulation been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Housing
		2	No	
cavity_wall_add_rep_12m	Has cavity wall insulation been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Housing
		2	No	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
solid_wall_add_rep_12m	Has solid wall insulation been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Housing
		2	No	
underfloor_add_rep_12m	Has underfloor insulation been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Housing
		2	No	
double_glazing_add_rep_12m	Have double or secondary glazed windows been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Housing
		2	No	
draughtproofing_add_rep_12m	Has draughtproofing on windows, doors and/or chimneys been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Housing
		2	No	
other_ins_add_rep_12m	Has other type of insulation been added or replaced in your home in the last 12 months?	1	Has been added or replaced in the last 12 months	Housing
		2	No	
mould_windows_bathroom	Do you have mould around the windows or in the bathroom?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	—	
		2	Minor	
		3	Substantial	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
mould_walls	Do you have mould on any walls?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	—	
		2	Minor	
		3	Substantial	
mould_furnishing	Do you have mould on furnishings?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	—	
		2	Minor	
		3	Substantial	
heating_difficulty_home	Do you feel your home is difficult to heat?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer / No response	
		-1	Don't know	
		0	No	
		1	Yes / You feel it is difficult to afford the fuel to heat your home	
		2	None of the above	
		3	Other reason	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
building_type	What type of accommodation do you live in?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Detached	
		2	Semi detached	
		3	Terraced (including end terrace)	
		4	In a purpose-built block of flats or tenement	
		5	Part of a converted or shared house (including bedsits)	
		6	In a commercial building (e.g. in an office building, hotel or over a shop)	
self_contained_accom	Is your accommodation self-contained?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes, all the rooms are behind a door that only my household can use	
		2	No	
no_bedrooms	How many of the rooms in your accommodation are bedrooms?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
no_bathrooms_ bath_shower	How many bathrooms do you have with a plumbed in bath or shower?	-9 -3 -2 -1 0 1 2 3 4	Not applicable Prefer not to say No answer Don't know 0 1 2 3 4 or more	Housing
own_rent_house	Do you (or your household) own or rent this accommodation?	-9 -3 -2 -1 1 2 3 4 5	Not applicable Prefer not to say No answer Don't know Own it outright / buying it with a mortgage/loan Part own and part rent (shared ownership) Rent it privately (with or without housing benefit) Rent it from council (local authority) or housing associa- tion (with or without housing benefit) Live here rent free	Housing

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
building_age	Approximately when do you think your accommodation was built?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Before 1900	
		2	1900 to 1929	
		3	1930 to 1949	
		4	1950 to 1975	
		5	1976 to 1990	
		6	1991 to 2002	
		7	2003 onwards	
		8	Don't know	
summer_overheating	During the summer does it ever get uncomfortably hot inside your home?	-9	Not applicable	Housing
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes, in all rooms	
		2	Yes, in multiple rooms but not all rooms	
		3	Yes, in one room	
		4	No	
Environment				
house_condition	Do you have problems with condensation, damp or mould?	-2	No answer	Environment
		-1	Don't know	
		1	Yes	
		0	No	
private_garden_access	Does your household have access to a private garden?	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
shared_garden_access	Does your household have access to a shared garden?	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
balcony_terrace_access	Does your household have access to a private balcony, private small patio, or private roof terrace?	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
shared_terrace_access	Does your household have access to a shared roof terrace?	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
green_space_access	Does your household have access to a park, wood, or other green space your household can access within easy walking distance of your home?	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
no_garden_park_access	Household has no access in green spaces (all of the above).	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
private_garage_present	Private garage, driveway	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
space_car_park_present	Personal space in a car park	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
no_allocated_park_space	Car park or parking area without allocated space	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
controlled_street_parking	On street in a controlled parking zone (e.g. residents' parking zone)	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
street_parking	On street, where there are no restrictions	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
other_parking_service	Other parking service (please type in)	-9	Not applicable	Environment
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
Finance				
financial_status	How well do you manage financially?	-9	Not applicable	Finance
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Living comfortably	
		2	Doing alright	
		3	Just about getting by	
		4	Finding it quite difficult	
		5	Finding it very difficult	
annual_income	Select option that corresponds with your recent gross household annual income?	-9	Not applicable	Finance
		-3	Prefer not to say / Prefer not to answer	
		-2	No answer	
		-1	Don't know	
		1	Below £10,000	
		2	£10,001 to £20,000	
		3	£20,001 to £30,000	
		4	£30,001 to £40,000	
		5	£40,001 to £50,000	
		6	£50,001 to £60,000	
		7	£60,001 to £70,000	
		8	£70,001 to £80,000	
		9	£80,001 to £90,000	
		10	£90,001 to £100,000	
		11	Above £100,000	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
payment_method_e1	Which of these methods do you use to pay for your energy? - electricity	-9 -3 -2 -1 1 2 3 4	Not applicable Prefer not to say No answer Don't know Direct debit (including online direct debit) Payment on receipt of bill (by post, telephone, online or at bank/post office) Pre-payment meter / Included in rent Other / Not applicable / no mains gas	Finance
payment_method_gas	Which of these methods do you use to pay for your energy? - gas	-9 -3 -2 -1 1 2 3 4	Not applicable Prefer not to say No answer Don't know Direct debit (including online direct debit) Payment on receipt of bill (by post, telephone, online or at bank/post office) Pre-payment meter / Included in rent Other / Not applicable / no mains gas	Finance

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
heating_ affordability	Do you feel it is difficult to afford the fuel to heat your home?	-9	Not applicable	Finance
		-3	Prefer not to say	
		-2	No answer / No response	
		-1	Don't know	
		0	No	
		1	Yes / You feel it is difficult to afford the fuel to heat your home	
		2	None of the above	
		3	Other reason	
heat_fuel_cost_ affordability	How easy or difficult is it for you to meet your heating/fuel costs?	1	Very easy	Finance
		2	Fairly easy	
		3	Neither easy nor difficult	
		4	Fairly difficult	
		5	Very difficult	
		6	Don't know	
		7	No response	
time_ dependence_ energy_price	Does the price you pay for your electricity depend on the time of day it is used? (for example, cheaper overnight or when there is less demand)	-9	Not applicable	Finance
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Don't know	
		Comfort		
thermal_comfort	During the cold winter weather, can you normally keep comfortably warm in your living room?	-9	Not applicable	Comfort
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
no_comfort_ answer	Prefer not to say, why not warm.	-9	Not applicable	Comfort
		-3	Prefer not to say	
		-2	No answer / No response	
		-1	Don't know	
		0	No	
		1	Yes / You feel it is difficult to afford the fuel to heat your home	
		2	None of the above	
		3	Other reason	
none_above_ comfort	None of the above applies for not being warm.	-9	Not applicable	Comfort
		-3	Prefer not to say	
		-2	No answer / No response	
		-1	Don't know	
		0	No	
		1	Yes / You feel it is difficult to afford the fuel to heat your home	
		2	None of the above	
		3	Other reason	
other_reason_ comfort	Other reason for not being warm.	-9	Not applicable	Comfort
		-3	Prefer not to say	
		-2	No answer / No response	
		-1	Don't know	
		0	No	
		1	Yes / You feel it is difficult to afford the fuel to heat your home	
		2	None of the above	
		3	Other reason	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
leave_home_to_warm	During this winter, do you leave your home to go to heated spaces elsewhere with the main purpose of keeping warm?	1	Daily	Comfort
		2	Most days	
		3	Rarely	
		4	Never	
		5	Don't know	
		6	Prefer not to say	
		7	No response	
Health				
life_satisfaction	Overall, how satisfied are you with your life nowadays, where 0 is 'not at all satisfied' and 10 is 'completely satisfied'?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0 not at all satisfied	
		1	1	
		2	2	
		3	3	
		4	4	
		5	5	
		6	6	
		7	7	
		8	8	
		9	9	
10	10 completely satisfied			

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
life_worthwhile_score	Overall, to what extent do you feel that the things you do in your life are worthwhile, where 0 is 'not at all worthwhile' and 10 is 'completely worthwhile'?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	0 not at all satisfied	
		1	1	
		2	2	
		3	3	
		4	4	
		5	5	
		6	6	
		7	7	
		8	8	
		9	9	
		10	10 completely satisfied	
health_condition	How is your health in general? Is it..	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Very good	
		2	Good	
		3	Fair	
		4	Bad	
		5	Very bad	
		6	Prefer not to say	
longterm_condition	Do you have any physical or mental health conditions or illnesses lasting or expected to last for 12 months or more?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Prefer not to say	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
vision	Is your vision fine?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
hearing	Is your hearing fine?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
learning_difficulty	Do you have learning difficulties?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
mobility	Are you mobile?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
breathing_problems	Do you have breathing problems?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
heart_disease	Do you have any heart disease?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
mental_health_probs	Do you have mental health problems?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
other_chronic_illness	Do you have any other chronic illness?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
dk_illness	Do you have any other chronic illness? - Don't know	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
preferenosay_illness	Do you have any other chronic illness? - Prefer not to say	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
hh_others_ longterm_ condition	Does anyone else in the household have any physical or mental health conditions or illnesses lasting or expected to last for 12 months or more?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		1	Yes	
		2	No	
		3	Prefer not to say	
		4	Not applicable (living alone)	
hh_others_ vision	Do other household members have any issues with their vision?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
hh_others_ hearing	Do other household members have any issues with their hearing?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
hh_others_ learning_ difficulty	Do other household members have any issues with learning?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
hh_others_ mobility	Do other household members have any issues with their vision?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
hh_others_breathing_problems	Do other household members have any issues with their breathing?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
hh_others_heart_disease	Do other household members have any heart disease?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
hh_others_mental_probs	Do other household members have any mental health problems?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
hh_others_other_chronic_illness	Do other household members have any other chronic illness?	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	
dk_hh_others_illness	Do other household members have any other chronic illness? - Don't know	-9	Not applicable	Health
		-3	Prefer not to say	
		-2	No answer	
		-1	Don't know	
		0	No	
		1	Yes	

Table 1 continued

Variable name	Concept	Possible values	Value meanings	Domain
prefernosay_hh_others_illness	Do other household members have any other chronic illness? - Prefer not to say	-9 -3 -2 -1 0 1	Not applicable Prefer not to say No answer Don't know No Yes	Health